# CH9 Partial Differential Equations

Sunday, April 12, 2026 4:50 PM

## 9.1 Introduction

PDE $\varphi(x,y)$ = dep. variable

$\frac{\partial \varphi}{\partial x}$ = partial derivative

$\frac{\partial^2 \varphi}{\partial x \partial y}$ = mixe PD

$\frac{\partial}{\partial x}$ is linear $\Leftrightarrow$ $\frac{\partial}{\partial x}\left[a\varphi(x,y)+b\psi(x,y)\right] = a\frac{\partial \varphi}{\partial x} + b\frac{\partial \psi}{\partial x}$

$\mathcal{L}$ contains partial derivative, multiple of $\varphi$ $\Rightarrow$ Linear

$\mathcal{L}\varphi(x,y) = F(x,y)$ ; $F(x,y) \equiv$ Source term $\begin{cases} \text{zero (Homogenous)} \\ \text{Nonzero (Inhomg.)} \end{cases}$

Separation of Varia

Green's Function

# Examples of PDE

Monday, April 13, 2026 3:09 PM

1) Laplace Equation $\nabla^2\psi = 0$ (Linear, homogenous)

2) Poisson " $\nabla^2\psi = f(\vec{r})$ (Linear, inhomog.)

3) Euler's eqn $\frac{\partial \vec{u}}{\partial t} + \vec{u}\cdot\vec{\nabla}\vec{u} = -\frac{\vec{\nabla}p}{\rho}$ (Nonlinear, inhomog.)

4) Helmholtz eqn, $\nabla^2\psi \pm k^2\psi = 0$

5) Time-dep- diffusion $\nabla^2\psi = \frac{1}{a^2}\frac{\partial\psi}{\partial t}$

6) " " Sch. Eqn $-\frac{\hbar^2}{2m}\nabla^2\psi + V\psi = i\hbar\frac{\partial\psi}{\partial t} \Rightarrow$ Time-ind $-\frac{\hbar^2}{2m}\nabla^2\psi + V\psi = E\psi$

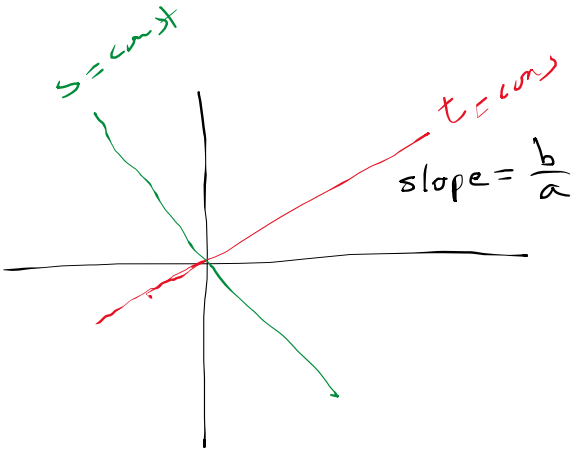
7) Time-dep Wave eqn $\nabla^2\psi = \frac{1}{c^2}\frac{\partial^2\psi}{\partial t^2}$

8) Klein-Gordon eq $\partial^2\psi = -\mu^2\psi$

# 9.2 1st Order PDE

Monday, April 13, 2026 3:19 PM

$$\mathcal{L}\varphi = a\frac{\partial \varphi}{\partial x} + b\frac{\partial \varphi}{\partial y} = 0$$




$s = ax + by$

$t = bx - ay$

$\varphi(x,y) = \hat{\varphi}(s,t)$

$$\frac{\partial \varphi}{\partial x} = \frac{\partial \hat{\varphi}}{\partial s}\frac{\partial s}{\partial x} + \frac{\partial \hat{\varphi}}{\partial t}\frac{\partial t}{\partial x} = a\frac{\partial \hat{\varphi}}{\partial s} + b\frac{\partial \hat{\varphi}}{\partial t}$$

$$\frac{\partial \varphi}{\partial y} = b\frac{\partial \hat{\varphi}}{\partial s} - a\frac{\partial \hat{\varphi}}{\partial t}$$

$$\Rightarrow \mathcal{L}\varphi = (a^2 + b^2)\frac{\partial \hat{\varphi}}{\partial s} = 0 \quad \Rightarrow \quad \hat{\varphi} = f(t) = f(bx - ay)$$

$t = bx - ay$ characteristic

# 9.3 2nd order PDE

Monday, April 13, 2026 3:27 PM

Ex $a^2 \frac{\partial^2 \psi}{\partial x^2} - b^2 \frac{\partial^2 \psi}{\partial y^2} = 0$ $\sin(kx \pm \omega t)$

$$\left(a\frac{\partial}{\partial x} + b\frac{\partial}{\partial y}\right)\left(a\frac{\partial}{\partial x} - b\frac{\partial}{\partial y}\right)\psi = 0$$

$$a\frac{\partial \psi}{\partial x} + b\frac{\partial \psi}{\partial y} = 0 \quad \text{or} \quad a\frac{\partial \psi}{\partial x} - b\frac{\partial \psi}{\partial y} = 0$$

$\Rightarrow \varphi_1(x,t) = f(bx - ay)$ or $\varphi_2(x,y) = g(bx + ay)$

Ex $a^2 \frac{\partial^2 \varphi}{\partial x^2} + b^2 \frac{\partial^2 \varphi}{\partial y^2} = 0$

$\left(a\frac{\partial}{\partial x} + ib\frac{\partial}{\partial y}\right)\left(a\frac{\partial}{\partial x} - ib\frac{\partial}{\partial y}\right)\varphi = 0 \Rightarrow$ Complex characteristics

# 2nd PDE

Monday, April 13, 2026 3:36 PM

$$\mathcal{L} = a\frac{\partial^2\varphi}{\partial x^2} + 2b\frac{\partial^2\varphi}{\partial x\partial y} + c\frac{\partial^2\varphi}{\partial y^2} = \left(\frac{b+\sqrt{b^2-ac}}{c}\frac{\partial}{\partial x} + \frac{1}{c}\frac{\partial}{\partial y}\right)\left(\frac{b-\sqrt{b^2-ac}}{c}\frac{\partial}{\partial x} + \frac{1}{c}\frac{\partial}{\partial y}\right)$$

Discr. $\equiv b^2 - ac > 0 \Rightarrow$ real char. — Hyperbolic

$< 0 \Rightarrow$ Complex " — Elliptic

$= 0 \Rightarrow$ one char. — Parabolic $\left(a\frac{\partial\varphi}{\partial x} = \frac{\partial^2\varphi}{\partial y^2}\right)$

Ex

| | | |
|---|---|---|
| Laplace | $\nabla^2\psi = 0$ | elliptic |
| Wave Eqn | $\nabla^2\psi = \frac{1}{c^2}\frac{\partial^2\psi}{\partial t^2}$ | hyperbolic |
| Diff Eq | $a\frac{\partial\psi}{\partial t} = \nabla^2\psi$ | parabolic |

# Boundary Conditions

Monday, April 13, 2026 3:47 PM

1. Cauchy BC: $\psi$, $\frac{\partial \psi}{\partial n}$ (ex: $\psi \equiv$ potential, $\frac{\partial \psi}{\partial n} \equiv$ electric field)

2. Dirichlet BC: $\psi$

3. Neuman BC: $\frac{\partial \psi}{\partial n}$

# 9.4 Separation of Variables

Monday, April 13, 2026 3:49 PM

## Cartesian Coord.

Helmholtz $\quad \frac{\partial^2\psi}{\partial x^2} + \frac{\partial^2\psi}{\partial y^2} + \frac{\partial^2\psi}{\partial z^2} + k^2\psi = 0$

Let $\psi(x,y,z) = X(x)\,Y(y)\,Z(z)$

$$YZ\frac{d^2X}{dx^2} + XZ\frac{d^2Y}{dy^2} + XY\frac{d^2Z}{dz^2} + k^2XYZ = 0$$

$$\frac{1}{X}\frac{d^2X}{dx^2} + \frac{1}{Y}\frac{d^2Y}{dy^2} + \frac{1}{Z}\frac{d^2Z}{dz^2} + k^2 = 0$$

$k^2 = l^2 + m^2 + n^2$

$-l^2$ $\qquad -m^2$ $\qquad -n^2$

$X'' = -l^2X \qquad Y'' = -m^2Y \qquad Z'' = -n^2Z$

$$\psi_{lmn}(x,y,z) = X_l(x)\, Y_m(y)\, Z_n(z)$$

General Solution $$\Psi(x,y,z) = \sum_{l,m} a_{lm} X_l(x) Y_m(y) Z_{lm}(z)$$

$$= \sum_{l,m} a_{lm} \psi_{lmn}(x,y,z)$$

Ex 9.4.1

Monday, April 13, 2026 3:58 PM

Dirichlet BC: $\psi = 0$ on all boundaries except $z = L$ ($\psi(z=L) = V$)

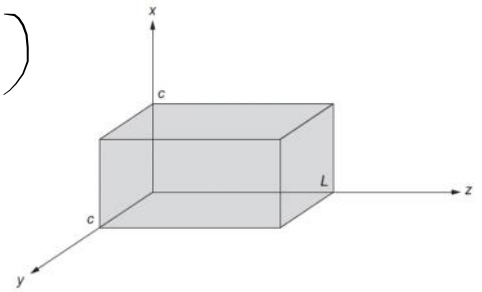


FIGURE 9.1 Parallelepiped for solution of Laplace equation.

$X'' = -l^2 X \Rightarrow X = A \sin lx + B \cos lx$

$X(0) = 0$, $X(c) = 0 \Rightarrow lc = \lambda\pi$ ; $\lambda = 1, 2, \ldots$

$X_\lambda(x) = \sin\left(\lambda \frac{\pi}{c} x\right)$

$Y'' = -m^2 Y \Rightarrow Y_\mu(y) = \sin\left(\mu \frac{\pi}{c} y\right)$ ; $\mu = 1, 2, \ldots$

$Z'' = -n^2 Z \Rightarrow Z'' = \frac{\pi^2}{c^2}(\lambda^2 + \mu^2) Z$

$Z'' = \rho^2 Z$

$n^2 = -(m^2 + l^2) = -\frac{\pi^2}{c^2}(\lambda^2 + \mu^2)$

$\rho = \rho_{\lambda\mu}$

$$Z_{\lambda\mu} = A e^{\rho z} + B e^{-\rho z}$$

$$Z(0) = 0 \Rightarrow Z_{\lambda\mu} = A' \sinh(\rho z)$$

$$Z(L) = V \Rightarrow Z_{\lambda\mu}(z) = V \frac{\sinh(\rho z)}{\sinh(\rho L)}$$

$$\Psi(x,y,z) = V \sum_{\lambda\mu} a_{\lambda\mu} \sin\left(\frac{\lambda\pi x}{c}\right) \sin\left(\frac{\mu\pi y}{c}\right) \frac{\sinh(\rho z)}{\sinh(\rho L)}$$

$$\Psi(x,y,L) = V = V \sum_{\lambda\mu} a_{\lambda\mu} \sin\left(\frac{\lambda\pi x}{c}\right) \sin\left(\frac{\mu\pi y}{c}\right)$$

$$1 = \left[\sum_{\lambda} b_\lambda \sin\left(\frac{\lambda\pi x}{c}\right)\right]^2$$

Symmetry $\Rightarrow a_{\lambda\mu} = b_\lambda b_\mu$

$$1 = \sum_{\lambda} b_\lambda \sin\left(\frac{\lambda \pi x}{c}\right)$$

$$b_\lambda = \frac{\int_0^c \sin\left(\frac{\lambda \pi x}{c}\right) dx}{\int_0^c \left[\sin\left(\frac{\lambda \pi x}{c}\right)\right]^2 dx} = \begin{cases} \frac{4}{\lambda \pi}, & \lambda \text{ odd} \\ 0, & \lambda \text{ even} \end{cases}$$

$$\Psi(x,y,z) = V \sum_{\lambda \mu} b_\lambda b_\mu \sin\left(\frac{\lambda \pi x}{c}\right) \sin\left(\frac{\mu \pi y}{c}\right) \frac{\sinh(\beta z)}{\sinh(\beta L)}$$

# Cylindrical Coord.

Wednesday, April 15, 2026 3:01 PM

$\psi = \psi(\rho, \varphi, z)$ $\quad \nabla^2\psi + k^2\psi = 0$

$$\frac{1}{\rho}\frac{\partial}{\partial\rho}\left(\rho\frac{\partial\psi}{\partial\rho}\right) + \frac{1}{\rho^2}\frac{\partial^2\psi}{\partial\varphi^2} + \frac{\partial^2\psi}{\partial z^2} + k^2\psi = 0$$

$\psi = P(\rho)\,\Phi(\varphi)\,Z(z)$

$$\frac{1}{\rho P}\frac{d}{d\rho}\left(\rho\frac{dP}{d\rho}\right) + \frac{1}{\rho^2\Phi}\frac{d^2\Phi}{d\varphi^2} + \frac{1}{Z}\frac{d^2Z}{dz^2} + k^2 = 0$$

let $\frac{1}{Z}\frac{d^2Z}{dz^2} = l^2$ $\qquad n^2 = l^2 + k^2$

mult. by $\rho^2$

$$\frac{\rho}{P}\frac{d}{d\rho}\left(\rho\frac{dP}{d\rho}\right) + \frac{1}{\Phi}\frac{d^2\Phi}{d\varphi^2} + n^2\rho^2 = 0 \quad \Rightarrow \quad \rho\frac{d}{d\rho}\left(\rho\frac{dP}{d\rho}\right) + (n^2\rho^2 - m^2)P = 0$$

$\Phi'' = -m^2\Phi$

$\psi_{lm}(\rho, \varphi, z) = P_{lm}(\rho)\,\Phi_m(\varphi)\,Z_l(z)$

$\Psi = \sum_{lm} a_{lm}\psi_{lm}$

Ex 9.4.2

Wednesday, April 15, 2026 3:12 PM

$-\nabla^2\psi = \lambda\psi$ Ground state soln.

Dirichlet : $\psi = 0$ at $\rho = R$, $z = \pm \frac{L}{2}$

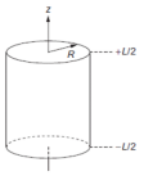



FIGURE 9.2 Cylindrical region for solution of the Helmholtz equation.

Helm. $\lambda = k^2$

$Z'' = l^2 Z$

$Z = A e^{lz} + B e^{-lz}$

BCX $l = i\omega$

$Z = A e^{i\omega z} + B e^{-i\omega z} \Rightarrow$

$Z = \cos\left(\frac{\pi}{L} z\right)$

$\omega = \frac{\pi}{L} \Rightarrow l^2 = -\frac{\pi^2}{L^2}$ $\Rightarrow n^2 = \lambda + l^2$ $\Rightarrow$ $\rho^2 P'' + \rho P' + n^2 \rho^2 P = 0$

$P = J_0(n\rho)$

$\Phi'' = -m^2 \Phi$

$\Phi = A' e^{im\phi} + B' e^{-im\phi}$

symmetry $\Rightarrow$ $m = 0, 1, 2, \ldots$

$\Phi = 1$

$P(S=R)=0 \Rightarrow J_0(nR)=0$

$nR=\alpha$ (smallest root)

$P=J_0\left(\frac{\alpha}{R}S\right)$

$\psi(S,\varphi,z)=J_0\left(\frac{\alpha\rho}{R}\right)\cos\left(\frac{\pi}{L}z\right)$

$\lambda=n^2-l^2=\frac{\alpha^2}{R^2}+\frac{\pi^2}{L^2}$

# Sph. Polar Coord.

Wednesday, April 15, 2026 3:39 PM

$$\frac{1}{r^2\sin\theta}\left[\sin\theta\frac{\partial}{\partial r}\left(r^2\frac{\partial\psi}{\partial r}\right)+\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial\psi}{\partial\theta}\right)+\frac{1}{\sin\theta}\frac{\partial^2\psi}{\partial\varphi^2}\right]=-k^2\psi$$

$$\frac{1}{\Phi}\Phi''=-m^2$$

$$\psi=R(r)\,\Theta(\theta)\,\Phi(\varphi)$$

$$\frac{1}{R r^2}\frac{d}{dr}\left(r^2\frac{dR}{dr}\right)+\frac{1}{\Theta r^2\sin\theta}\frac{d}{d\theta}\left(\sin\theta\frac{d\Theta}{d\theta}\right)+\frac{1}{\Phi r^2\sin^2\theta}\frac{d^2\Phi}{d\varphi^2}+k^2=0$$

$$\frac{1}{R r^2}\frac{d}{dr}\left(r^2\frac{dR}{dr}\right)+\frac{1}{\Theta r^2\sin\theta}\frac{d}{d\theta}\left(\sin\theta\frac{d\Theta}{d\theta}\right)-\frac{m^2}{r^2\sin^2\theta}+k^2=0$$

$$\underbrace{\frac{1}{R}\frac{d}{dr}\left(r^2\frac{dR}{dr}\right)+k^2r^2}_{\lambda}+\underbrace{\frac{1}{\Theta\sin\theta}\frac{d}{d\theta}\left(\sin\theta\frac{d\Theta}{d\theta}\right)-\frac{m^2}{\sin^2\theta}}_{-\lambda}=0$$

$$\frac{1}{\sin\theta}\frac{d}{d\theta}\left(\sin\theta\frac{d\Theta}{d\theta}\right)-\frac{m^2}{\sin^2\theta}\Theta+\lambda\Theta=0$$

$P(\cos\theta)=\Theta(\theta)$

$x=\cos\theta$ "Hw" $\Rightarrow$ $(1-x^2)P''-2xP'-\frac{m^2}{1-x^2}P+\lambda=0$

Ass. Leg. Eq.

$\lambda=l(l+1)$

$$\frac{1}{r^2}\frac{d}{dr}\left(r^2\frac{dR}{dr}\right)+k^2R-\frac{\lambda R}{r^2}=0$$

$$r^2R''+2rR'+\left[k^2r^2-l(l+1)\right]R=0$$

$k=0$ "Laplace's eqn" $\Rightarrow$ $r^2R''+2rR'-l(l+1)R=0$

Euler's ODE $\Rightarrow$ $R(r)=Ar^l+Br^{-(l+1)}$

$$r^2 R'' + 2rR' + [k^2 r^2 - l(l+1)] R = 0$$

Nonzero k

$$R(r) = \frac{Z(kr)}{(kr)^{1/2}}$$

Hw

$$x^2 Z'' + xZ' + \left[x^2 - \left(l + \tfrac{1}{2}\right)^2\right] Z = 0$$

$$Z(x) = J_{l+\frac{1}{2}}(x)$$

$$R(x) = j_l(x)$$

Sph. Bessel $\Rightarrow R(r) = A\, j_l(kr) + B\, y_l(kr)$

$$\psi(r,\theta,\varphi) = \sum_{lm} \left[A_{lm}\, j_l(kr) + B_{lm}\, y_l(kr)\right] P_l^m(\cos\theta)\left(A'_{lm} \sin m\varphi + B'_{lm} \cos m\varphi\right)$$

Ex 9.4.4

Wednesday, April 15, 2026 4:06 PM

$$\frac{d\psi}{d\hat{n}} = -V_0 \cos\theta$$

$$\frac{d\psi}{d\hat{n}} = -\frac{\partial \psi}{\partial r}$$

$$-V_0 \cos\theta = -\sum_{l,m} A_{lm}\, l\, a^{l-1}\, P_l^m(\cos\theta)\,(A'_{lm}\sin m\varphi + B'_{lm}\cos m\varphi)$$

$m = 0$

$$-V_0 \cos\theta = -\sum_l A_l\, l\, a^{l-1}\, P_l(\cos\theta)$$

$$V_0\, \delta_{l1} = A_1$$

$$\Rightarrow \psi = V r\cos\theta + A_0 = Vz + A_0$$

$$\oint \vec{E}\cdot d\vec{A} = \int_S \cos\theta\, d\sigma = \int_0^{\pi} \cos\theta\, d\theta \int_0^{2\pi} d\varphi = 0$$

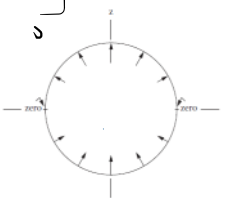



FIGURE 9.3 Arrows indicate sign and relative magnitude of the (inward) normal derivative of the electrostatic potential on a spherical surface (boundary condition for Example 9.4.4).

Table 9.2 Solutions of PDEs in Spherical Polar Coordinates[a]

$$\psi = \sum_{l,m} f_l(r) P_l^m(\cos\theta) \left\{ \begin{array}{c} a_{lm}\cos m\varphi + b_{lm}\sin m\varphi) \\ \text{or} \\ c_{lm} e^{im\varphi} \end{array} \right\}$$

| | |
|---|---|
| $\nabla^2\psi = 0$ | $f_l(r) = r^l,\ r^{-l-1}$ |
| $\nabla^2\psi + k^2\psi = 0$ | $f_l(r) = j_l(kr),\ y_l(kr)$ |
| $\nabla^2\psi - k^2\psi = 0$ | $f_l(r) = i_l(kr),\ k_l(kr)$ |

[a] For $i_l$, $j_l$, $k_l$, $y_l$, see Chapter 14; for $P_l^m$, see Chapters 15 and 16.

Table 9.3 Solutions of PDEs in Circular Cylindrical Coordinates[a]

$$\psi = \sum_{m,\alpha} f_{m\alpha}(\rho) g_\alpha(z) \left\{ \begin{array}{c} a_{m\alpha}\cos m\varphi + b_{m\alpha}\sin m\varphi) \\ \text{or} \\ c_{m\alpha} e^{im\varphi} \end{array} \right\}$$

| | | | |
|---|---|---|---|
| $\nabla^2\psi = 0$ | | $f_{m\alpha}(\rho) = J_m(\alpha\rho), Y_m(\alpha\rho)$ | $g_\alpha(z) = e^{\alpha z},\ e^{-\alpha z}$ |
| | or | $f_{m\alpha}(\rho) = I_m(\alpha\rho), K_m(\alpha\rho)$ | $g_\alpha(z) = \sin(\alpha z), \cos(\alpha z)$ or $e^{i\alpha z}$ |
| | or | $f_{m\alpha}(\rho) = \rho^m,\ \rho^{-m}$ | $g_\alpha(z) = 1$ |
| $\nabla^2\psi + \lambda\psi = 0$ | | $f_{m\alpha}(\rho) = J_m(\alpha\rho), Y_m(\alpha\rho)$ | |
| | | if $\beta^2 = \alpha^2 - \lambda > 0$, | $g_\alpha(z) = e^{\beta z},\ e^{-\beta z}$ |
| | | if $\beta^2 = \lambda - \alpha^2 > 0$, | $g_\alpha(z) = \sin(\beta z), \cos(\beta z)$ or $e^{i\beta z}$ |
| | | if $\lambda = \alpha^2$, | $g_\alpha(z) = 1$ |
| | or | $f_{m\alpha}(\rho) = I_m(\alpha\rho), K_m(\alpha\rho)$ | |
| | | if $\beta^2 = -\lambda - \alpha^2 > 0$, | $g_\alpha(z) = e^{\beta z},\ e^{-\beta z}$ |
| | | if $\beta^2 = \lambda + \alpha^2 > 0$, | $g_\alpha(z) = \sin(\beta z), \cos(\beta z)$ or $e^{i\beta z}$ |
| | | if $\lambda = -\alpha^2$, | $g_\alpha(z) = 1$ |
| | or | $f_{m\alpha}(\rho) = \rho^m,\ \rho^{-m}$ | |
| | | if $\beta^2 = -\lambda > 0$, | $g_\alpha(z) = e^{\beta z},\ e^{-\beta z}$ |
| | | if $\beta^2 = \lambda > 0$, | $g_\alpha(z) = \sin(\beta z), \cos(\beta z)$ or $e^{i\beta z}$ |

[a] The parameter $\alpha$ can have any real values consistent with the boundary conditions. For $I_m$, $J_m$, $K_m$, $Y_m$, see Chapter 14.